Development of an Artificial Intelligence (AI) Chatbot to Support Surgeon–Patient Communication in Thyroid Cancer Care

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Introduction

- Patients struggle with post-surgery questions
- Delays in communication increase anxiety
- No scalable solution exists

OBJECTIVE: Reduce surgeon–patient communication delays

Why AI?

- Immediate responses
- Scalable support
- Handles repetitive patient questions

RAG vs Non-RAG

Non-RAG:

- Answers from memory
- Can miss details
- Risk of hallucination

RAG:

- Retrieves real medical info
- More complete answers
- Safer for patients

Methods

Data:
- 177 patient Q&A
- Verified by surgeon

Models:
- RAG chatbot
- 6 non-RAG models

Evaluation:
- LLM-as-judge
- Safety, completeness, correctness

Findings Overview

- Non-RAG models: ~8.4 avg
- Only one model approached 9.0
- RAG improves safety + completeness (key for patients)

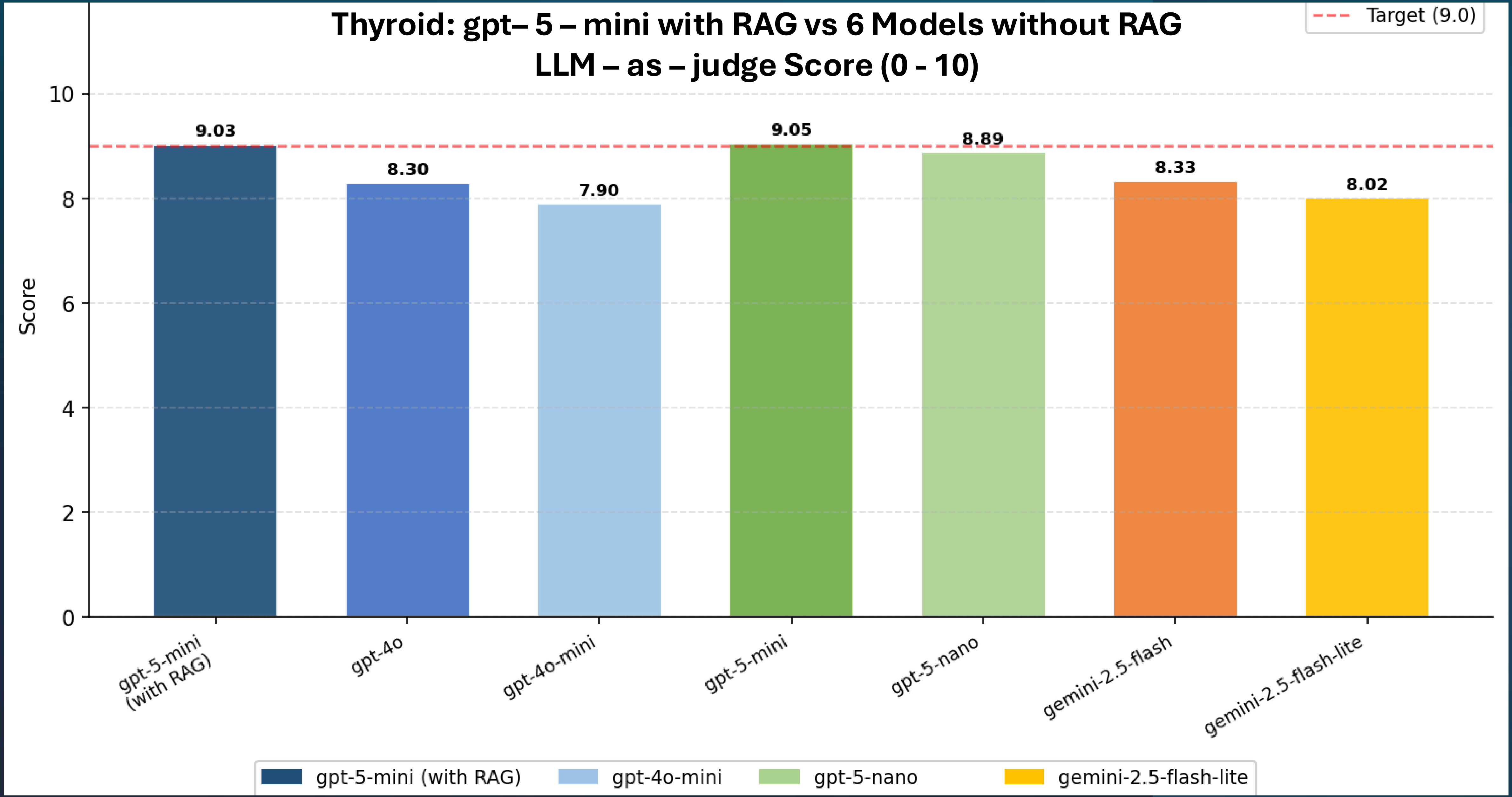

Figure 1. The RAG-enhanced GPT-5-mini (9.03) outperformed most standalone models.

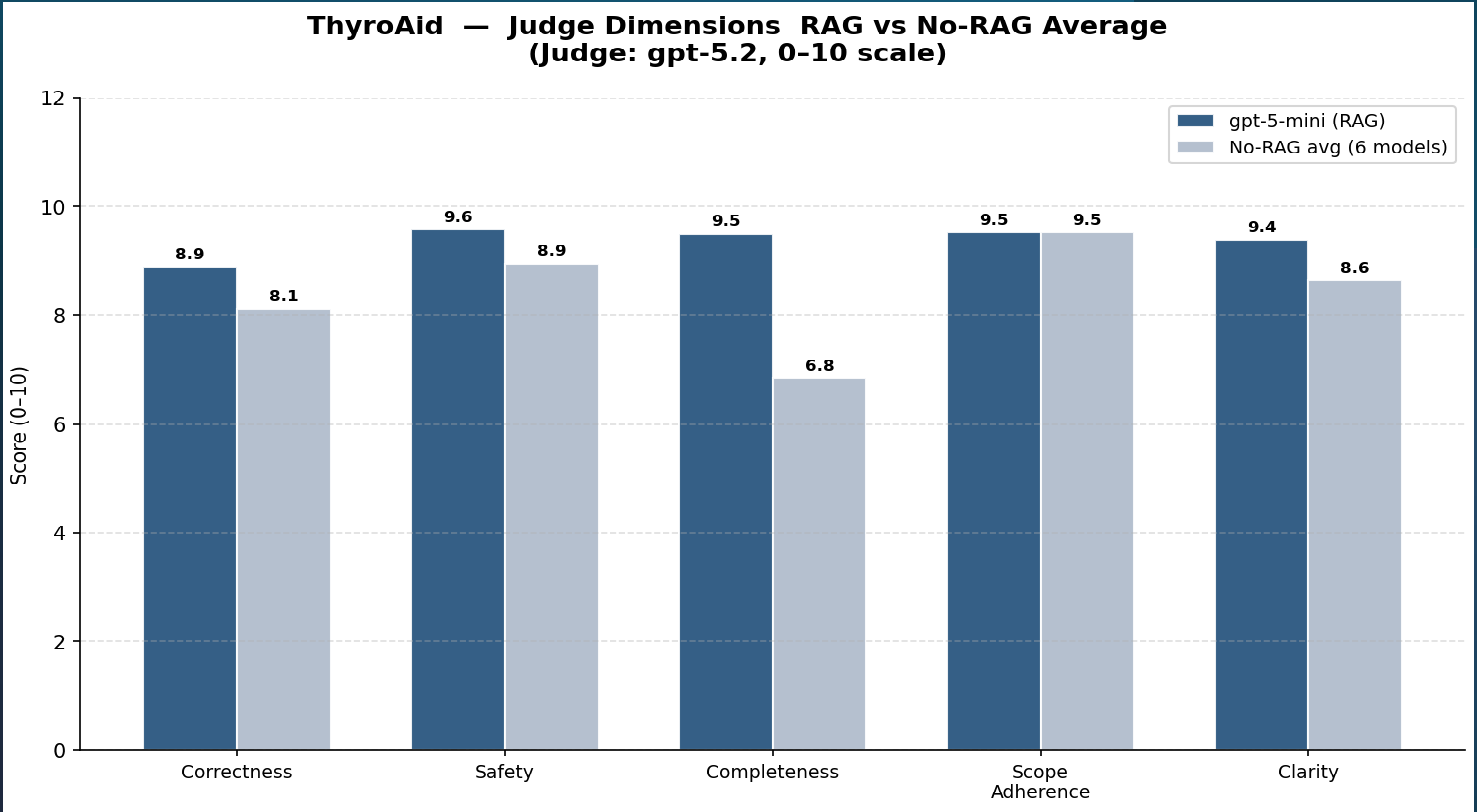

Figure 2. The RAG model outperformed the non-RAG average in safety, completeness, and correctness, showing improved answer quality in key clinical dimensions.

Conclusion

- RAG chatbot outperforms non-RAG models
- Achieves highest safety and completeness
- Suitable for patient-facing use

Next Steps

- Test with real patients
- Collect feedback (clarity, trust)
- Measure impact on communication delays

Any questions?